

Homework Assignment 3: Critical Systems by Diffusion, P_N and S_N

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Question 1: Critical radius of a reflected sphere

The diffusion equation

The two regions, and the solutions to Helmholtz on each

Every radius here is measured in mean free paths of the medium it sits in:

$$a = \Sigma_{t,C} R_C, \quad b = \Sigma_{t,R} R_R = a + d \quad (1)$$

the dimensionless core radius and the dimensionless outer radius of the reflector, whose width is the given d .

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{1}{r} \frac{d^2 u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2 u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0} u = 0 \quad (2)$$

from which the scalar flux is calculated to be (see slide 2 in Heizler's relevant presentation):

$$\phi_C(r) = A \frac{\sin(k_0(c_C) \Sigma_{t,C} r)}{\Sigma_{t,C} r}, \quad \phi_R(r) = F \frac{\sinh([b + z_0(c_R) - \Sigma_{t,R} r] / \nu_0(c_R))}{\Sigma_{t,R} r} \quad (3)$$

Derivation. The core has $c_C > 1$, so the coefficient is negative and the solution oscillates. Finiteness at the origin selects the sine. The reflector has $c_R \leq 1$ and decays, vanishing $z_0(c_R)$ mean free paths past b . The relaxation rate is $k_0(c_C) = \sqrt{(c_C - 1)/D_0(c_C)}$ per mean free path in the core and $1/\nu_0(c_R) = \sqrt{(1 - c_R)/D_0(c_R)}$ in the reflector. But remember! k_0 and ν_0 are Case's discrete eigenvalues on its two branches (for c smaller or greater than 1), following the identities: $c \arctan(k_0) = k_0$ for $c > 1$ and $\text{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ otherwise.

Matching, and the three criticality equations

The interface at $r = R_C$ carries a flux relation $\phi_R = \rho_{2/1} \phi_C$ and a current relation with jump factor $j_{2/1}$. The two continuous approximations set $\rho_{2/1} = j_{2/1} = 1$ and impose both on ϕ itself. The discontinuous one takes its two ratios from the plane two-region problem, where $u = r\phi$ plays the part of the plane flux, and so imposes them on u . Each set gives one transcendental equation for a , and the three cases (a), (b), (c) are the two equations filled with the parameters of Table 1.

(a) and (b): ϕ and the net current continuous. The conditions are

$$\phi_R(R_C) = \phi_C(R_C) \quad (4)$$

$$-\frac{D_R}{\Sigma_R} \left(\frac{d\phi_R(r)}{dr} \right)_{r=R_C} = -\frac{D_C}{\Sigma_C} \left(\frac{d\phi_C(r)}{dr} \right)_{r=R_C} \quad (5)$$

In optical variables that current is dimensionless on both sides, $D d/dr = D_0 d/d(\Sigma_t r)$, so $D_R = D_{0,R}$ and $D_C = D_{0,C}$ below are the dimensionless D_0 of Table 1. Under substitution of the scalar fluxes the two conditions turn to:

$$\frac{F}{b-d} \sinh\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right) = \frac{A}{a} \sin(k_0(c_C)a) \quad (6)$$

$$D_R F \left[\frac{\cosh\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right)}{\nu_0(c_R)(b-d)} + \frac{\sinh\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right)}{(b-d)^2} \right] = D_C A \left[\frac{\sin(k_0(c_C)a)}{a^2} - \frac{k_0(c_C) \cos(k_0(c_C)a)}{a} \right] \quad (7)$$

divide the two and get:

$$\cot(k_0 a) = \frac{1}{k_0 a} - \frac{D_R}{D_C k_0} \left[\frac{1}{\nu_0} \coth\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right) + \frac{1}{b-d} \right] \quad (8)$$

(c): the fixed jump conditions. The *fixed* jump conditions for the curvilinear coordinates under the spherical symmetry relation $\phi \rightarrow r\phi$ and Zimmerman's $\rho_{2/1} = \mu_0(c_C)/\mu_0(c_R)$, $j_{2/1} = 1$ are:

$$\mu_0(c_R) [r\phi_R(r)]_{r=R_C} = \mu_0(c_C) [r\phi_C(r)]_{r=R_C} \quad (9)$$

$$-\frac{D_R}{\Sigma_R} \left(\frac{d[r\phi_R(r)]}{dr} \right)_{r=R_C} = -\frac{D_C}{\Sigma_C} \left(\frac{d[r\phi_C(r)]}{dr} \right)_{r=R_C} \quad (10)$$

which under substitution of the scalar fluxes turn to:

$$\mu_0(c_R) \frac{F}{\Sigma_R} \sinh\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right) = \mu_0(c_C) \frac{A}{\Sigma_C} \sin(k_0(c_C)a) \quad (11)$$

$$-\frac{D_R F}{\Sigma_R \nu_0(c_R)} \cosh\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right) = \frac{j_{2/1} D_C A k_0(c_C)}{\Sigma_C} \cos(k_0(c_C)a) \quad (12)$$

divide the two and get:

$$\cot(k_0 a) = -\frac{\mu_0(c_C)}{\mu_0(c_R)} \frac{D_R/\nu_0}{D_C k_0} \coth\left(\frac{d+z_0(c_R)}{\nu_0(c_R)}\right) \quad (13)$$

writing $k_0 = k_0(c_C)$ and $\nu_0 = \nu_0(c_R)$. Notice that d is defined here such that $b-d = \Sigma_R R_C$. The code keeps the two conversions apart. The reflector enters through nothing but the optical combination $(d+z_0(c_R))/\nu_0(c_R)$, its thickness in its own diffusion lengths. Both equations run from $+\infty$ at $a \rightarrow 0$ to $-\infty$ at $a = \pi/k_0$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/k_0)$ and `brentq` finds it in one call.

So the three cases are

- (a) with $D_0 = \frac{1}{3}$, $z_0 = \frac{2}{3}$ (b) with the asymptotic D_0 , z_0 (c) with the asymptotic D_0 , z_0

The three approximations differ only in what they put into D_0 and z_0 :

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	1/3	$ 1 - c \nu_0^2$	$ 1 - c \nu_0^2$
$z_0(c)$	2/3	$0.710446 [1 + q(1 - c)^2]/c$	$0.710446 [1 + q(1 - c)^2]/c$
$\rho_{2/1} = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: The three approximations differ only in what they put into D_0 and z_0 . The discontinuous one also takes the flux ratio from Zimmerman.

Zimmerman's μ_0 . The discontinuity comes from Zimmerman, who writes the partial currents of the asymptotic mode as $j_{\pm} = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{\int_0^1 \mu \psi_{asym}(\mu) d\mu}{\frac{1}{2} \int_{-1}^1 \psi_{asym}(\mu) d\mu} = \frac{c \nu_0^2}{2} \ln\left(\frac{\nu_0^2}{\nu_0^2 - 1}\right) \quad (14)$$

where

$$\nu_0 = \nu_0(c) \text{ for } c < 1, \quad |\nu_0| = 1/k_0(c) \text{ for } c > 1 \quad (15)$$

the second branch being the analytic continuation through $\nu_0 = i/k_0$, which is real and equal to $\frac{c}{2k_0^2} \ln(1 + k_0^2)$. It returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam.

material	c	mfp (Σ_t^{-1})[cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H ₂ O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials according to their cross-section values from Assignment 1.

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.4195	9.879	7.968	5.677
Water	2	5.0855	4.7987	4.2459	8.649	7.267	5.034
Water	3	5.0153	4.7499	4.1906	8.296	7.048	4.840
Water	10	4.9814	4.7244	4.1620	8.129	6.935	4.741
Iron	1	5.4212	4.9575	4.2252	10.478	8.013	4.961
Iron	2	5.1056	4.7364	3.9170	8.753	6.988	3.952
Iron	3	4.9506	4.6237	3.7661	7.979	6.501	3.513
Iron	10	4.6790	4.4188	3.5037	6.737	5.674	2.829
Sodium	1	6.4397	5.7159	4.2207	17.562	12.281	4.945
Sodium	2	6.3134	5.6405	3.9087	16.549	11.801	3.927
Sodium	3	6.2505	5.6022	3.7544	16.059	11.563	3.480
Sodium	10	6.1315	5.5286	3.4701	15.159	11.113	2.748

Table 4: Pu-239 core, critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.4284	30.330	26.273	21.143
Water	2	6.9185	6.6731	6.1577	26.355	23.649	18.582
Water	3	6.8146	6.5937	6.0691	25.186	22.816	17.791
Water	10	6.7638	6.5520	6.0228	24.627	22.385	17.387
Iron	1	7.2811	6.8392	6.1246	30.720	25.460	18.284
Iron	2	6.8002	6.4625	5.6147	25.027	21.480	14.087
Iron	3	6.5545	6.2644	5.3543	22.411	19.565	12.217
Iron	10	6.1125	5.8970	4.8895	18.176	16.320	9.303
Sodium	1	8.4264	7.7951	6.1175	47.618	37.697	18.220
Sodium	2	8.2322	7.6564	5.6006	44.401	35.721	13.981
Sodium	3	8.1328	7.5844	5.3339	42.811	34.722	12.077
Sodium	10	7.9401	7.4425	4.8292	39.840	32.809	8.963

Table 5: U-235 core, same columns.

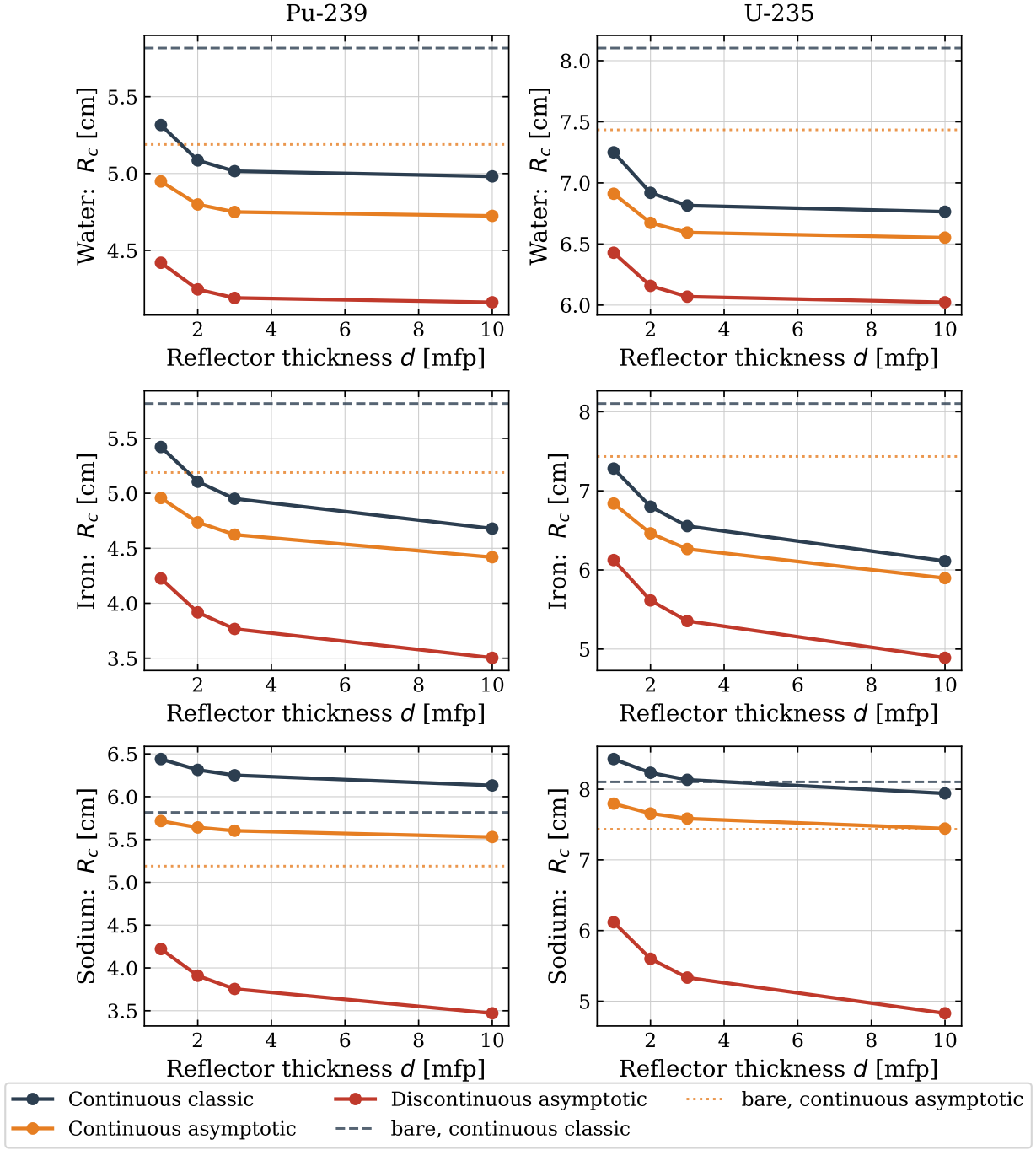


Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d = 3$ mfp, iron is still improving at $d = 10$ mfp and the sodium (a) and (b) curves sit above their bare lines throughout, while (c) does not.

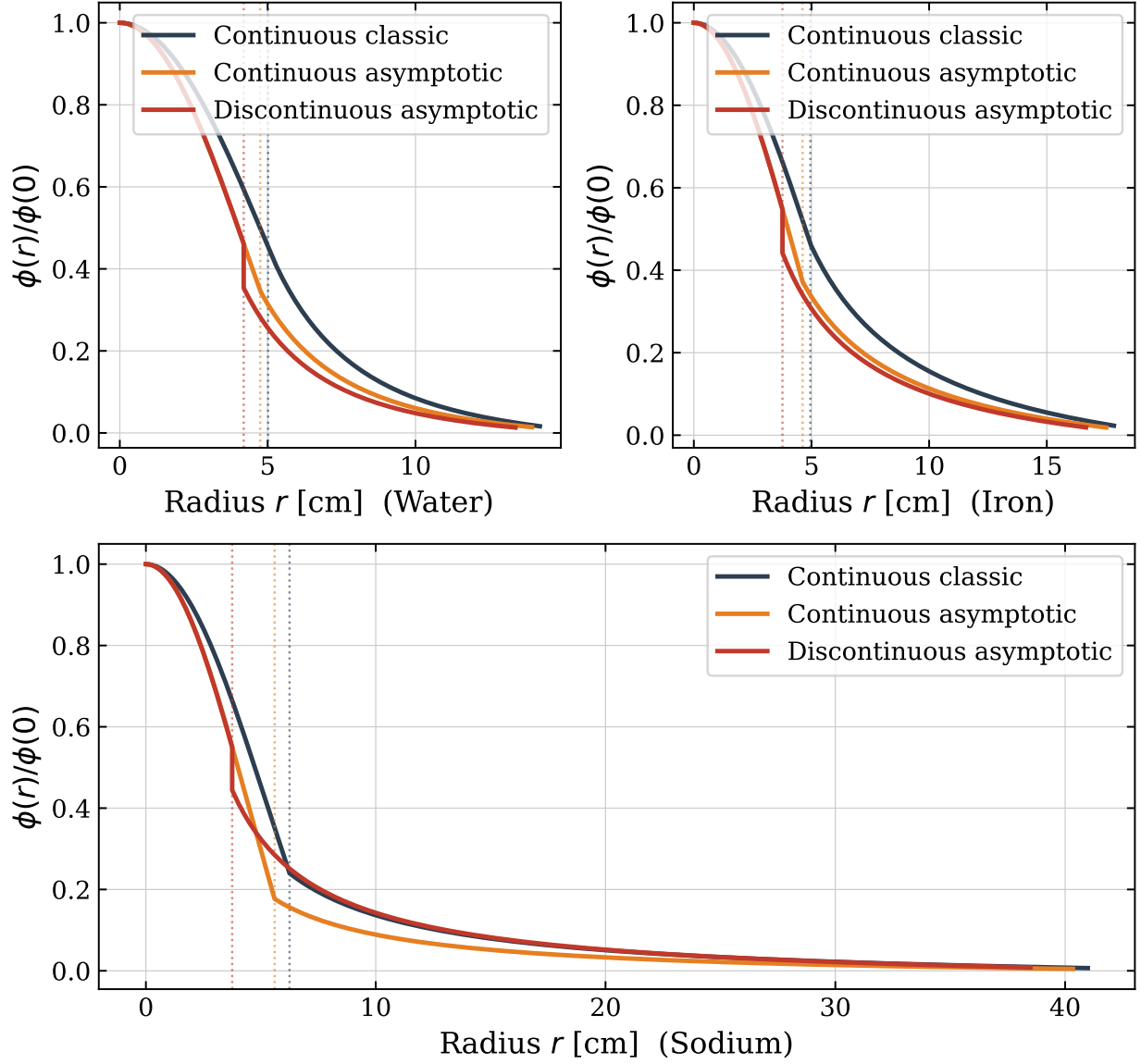


Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$. Dotted verticals mark each theory's interface. The discontinuous curve drops by the factor $\mu_0(c_R)/\mu_0(c_C)$ at the interface. Note the horizontal scale of the sodium panel, the wide one beneath: 3 mfp of sodium is 34.7 cm.

Question 2: Critical slab half-thickness by P_N

A homogeneous slab of total thickness a , reflective at the midplane $x = 0$ and vacuum at $x = a/2$, in mean free paths ($\Sigma_t = 1$), for $c \in \{1.2, 1.5, 1.8\}$ and $N \in \{1, 3, 5, 9\}$. The unknown is the critical half-thickness $a/2$ at which $k_{\text{eff}} = 1$ - the same quantity Question 3 tabulates for S_N , written the same way, so the two tables compare entry by entry.

The P_N equations

Questions 2–5 all solve the same one-speed slab equation, written out as scattering and fission kept apart, and only the fission source divided by k ,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi(x, \mu) = \frac{1}{2} \left[\Sigma_s \phi(x) + \frac{1}{k} \nu \Sigma_f \phi(x) \right] \quad (16)$$

Since each one of the questions 2-4 specify only c rather than giving Σ_s and $\nu \Sigma_f$ separately, taking the free parameter to be $\Sigma_s = 0$ causes $\nu \Sigma_f = c$ and puts the entire source under k , leaving an equation in c alone:

$$\mu \frac{\partial \psi}{\partial x} + \psi(x, \mu) = \frac{c}{2k} \phi_0(x), \quad \phi_n(x) = \int_{-1}^1 P_n(\mu) \psi(x, \mu) d\mu \quad (17)$$

Explaining why Σ_s can be chosen freely

This question fixes $\Sigma_t = 1$ and $c = (\Sigma_s + \nu \Sigma_f)/\Sigma_t$ and nothing else, so Σ_s is a free parameter: every split $\nu \Sigma_f = c - \Sigma_s$ with $0 \leq \Sigma_s \leq c$ realises the stated c , and they are different media. The split cannot change the quantity asked for, because at $k = 1$ the bracket collapses to $\frac{c}{2} \phi$ whatever it is, so the critical half-thickness is a function of c alone. It does change k at every other half-thickness - Σ_s is not divided by k - so c alone fixes the root of $k(a/2) - 1$ but not $k(a/2)$ itself.

Expanding $\psi(x, \mu) = \sum_{n' \geq 0} \frac{2n'+1}{2} \phi_{n'}(x) P_{n'}(\mu)$, multiplying (17) by $P_n(\mu)$ and integrating over μ turns the streaming term into a three-term coupling through the Legendre recurrence $(2n+1)\mu P_n = (n+1)P_{n+1} + nP_{n-1}$, while the collision term is diagonal and the isotropic source reaches only $n = 0$:

$$\frac{n+1}{2n+1} \frac{d\phi_{n+1}}{dx} + \frac{n}{2n+1} \frac{d\phi_{n-1}}{dx} + \phi_n(x) = \delta_{n0} \frac{c}{k} \phi_0(x), \quad n = 0, \dots, N \quad (18)$$

The system closes on itself once $\phi_{N+1} \equiv 0$ is imposed, which is the entire content of the P_N approximation (also notice that $\tilde{\mathbf{A}}_{\mathbf{0}, -1} = \mathbf{0}$ so this closure over the $N+1$ equations is handed for free). In vector form, with $\Phi = (\phi_0, \dots, \phi_N)^T$,

$$\tilde{\mathbf{A}} \frac{d\Phi}{dx} + \Sigma(k) \Phi = 0, \quad \tilde{A}_{n,n+1} = \frac{n+1}{2n+1}, \quad \tilde{A}_{n,n-1} = \frac{n}{2n+1}, \quad \Sigma_0(k) = \text{diag}\left(1 - \frac{c}{k}, 1, \dots, 1\right) \quad (19)$$

$\tilde{\mathbf{A}}$ is the Jacobi matrix of the Legendre recurrence truncated at order N , so its $N+1$ eigenvalues are exactly the roots of $P_{N+1}(\mu)$ - the Gauss-Legendre ordinates of S_{N+1} . That is the algebraic content of the claim in Question 3 that the two methods share their interior equations and differ only at the boundary.

Boundary conditions

At the midplane ($x = 0$) the flux is even in μ , so every odd moment vanishes:

$$\phi_{2m+1}(0) = 0, \quad m = 0, \dots, \frac{N-1}{2} \quad (20)$$

Equivalently $d\phi_{2m}/dx|_0 = 0$, which (18) then delivers for free (Notice that only one of the two families may be imposed or the problem is over-determined.)s

At $x = a/2$ no neutron enters, $\psi(a/2, \mu) = 0$ for $\mu < 0$, which $N + 1$ moments cannot satisfy pointwise. Marshak's choice is to annihilate the $\frac{N+1}{2}$ lowest odd moments of the incoming half-range,

$$\int_{-1}^0 \mu^{2m-1} \psi\left(\frac{a}{2}, \mu\right) d\mu = 0 \implies \sum_{n=0}^N c_{mn} \phi_n\left(\frac{a}{2}\right) = 0, \quad c_{mn} = \frac{2n+1}{2} \int_{-1}^0 \mu^{2m-1} P_n(\mu) d\mu \quad (21)$$

for $m = 1, \dots, \frac{N+1}{2}$. The coefficients follow from $P_n(-\mu) = (-1)^n P_n(\mu)$,

$$c_{mn} = (-1)^{n+1} \frac{2n+1}{2} \int_0^1 \mu^{2m-1} P_n(\mu) d\mu \quad (22)$$

whose integrand is a polynomial of degree at most $2N$, so an $(N + 1)$ -point Gauss-Legendre rule mapped to $[0, 1]$ evaluates it exactly rather than approximately as done in `pn.algebra.marshak_matrix`.

To sum up, (18) is $N + 1$ first-order equations and needs $N + 1$ conditions. (20) supplies $\frac{N+1}{2}$ and (21) the other $\frac{N+1}{2}$.

Box discretisation and the Bell & Glasstone k iteration

The scheme. Divide $[0, a/2]$ into J cells of width $\Delta x = a/(2J)$ with nodes $x_j = j\Delta x$. Integrating (19) over one cell and closing it with the midpoint rule - derivative by the difference across the cell, value by the average of the two faces -

$$\left. \frac{d\phi_n}{dx} \right|_{j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} - \phi_{n,j}}{\Delta x}, \quad \phi_{n,j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} + \phi_{n,j}}{2} \quad (23)$$

gives one $(N + 1)$ -row block per cell,

$$\left(\frac{\tilde{\mathbf{A}}}{\Delta x} + \frac{\mathbf{I}}{2} \right) \Phi_{j+1} + \left(\frac{\mathbf{I}}{2} - \frac{\tilde{\mathbf{A}}}{\Delta x} \right) \Phi_j = q_{j+\frac{1}{2}} \mathbf{e}_0, \quad j = 0, \dots, J-1 \quad (24)$$

The linear system. Stacking the $J + 1$ nodal vectors gives $(J + 1)(N + 1)$ unknowns. Equation (24) supplies $J(N + 1)$ rows, (20) supplies $\frac{N+1}{2}$ rows acting on Φ_0 and (21) the last $\frac{N+1}{2}$ acting on Φ_J . the count closes exactly. Ordered node by node the matrix is block-bidiagonal with two boundary blocks, i.e. banded with half-bandwidth $N + 1$, and one banded (or sparse) LU factorisation solves it. Call that matrix $\mathbf{L}$: it carries streaming, collision and both boundary conditions, and it does *not* depend on k .

The k update. Bell & Glasstone update k by the ratio of successive fission productions,

$$k^{(\ell+1)} = k^{(\ell)} \frac{\int_0^{a/2} \frac{c \Sigma_t}{k^{(\ell)}} \phi_0^{(\ell+1)} dx}{\int_0^{a/2} \frac{c \Sigma_t}{k^{(\ell)}} \phi_0^{(\ell)} dx} = k^{(\ell)} \frac{\int_0^{a/2} \phi_0^{(\ell+1)} dx}{\int_0^{a/2} \phi_0^{(\ell)} dx} \quad (25)$$

The middle form is what the method actually compares: the fission source (24) that $\phi_0^{(\ell+1)}$ would build against the one that produced it, so it is an exact ratio of sources and not of fluxes that happen to resemble them. The iterate is then rescaled by $\max_j \phi_{0,j}$, which has no effect on (25) - it only keeps the vector from drifting towards 0 or ∞ as k^ℓ compounds - and the loop stops on $|k^{(\ell+1)} - k^{(\ell)}| < \epsilon_k$. Convergence is linear at the dominance ratio of the two leading eigenvalues, which for a bare slab is well separated, and the starting guess

$$\phi_0^{(0)}(x) = \cos\left(\frac{\pi x}{2(a/2 + 0.7104)}\right), \quad k^{(0)} = 1 \quad (26)$$

is the diffusion fundamental mode with the asymptotic extrapolation distance, already almost the answer.

The outer loop. Dividing the fission source by k makes a solution exist at every thickness, so criticality is a root search on

$$f(a/2) = k(a/2) - 1 \quad (27)$$

Algorithm 1: Critical half-thickness by the box P_N scheme - homework3.pn.box

Input: secondaries c , order N , cells J , initial guess $(a/2)_0$
Output: $(a/2)_{\text{crit}}$ such that $k = 1$

```
1 Function BoxSystem( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ):  
    /* Setup -- once per  $(a/2, N)$ , outside every loop */  
2     $\Delta x \leftarrow (a/2)/J$ ;  
3    assemble  $\mathbf{L}$  of order  $(J+1)(N+1)$ :  $\frac{N+1}{2}$  symmetry rows (20) on  $\Phi_0$ .  $J$  cell blocks (24).  
     $\frac{N+1}{2}$  Marshak rows (21) on  $\Phi_J$ ;  
4    return splu( $\mathbf{L}$ ) ; // factorise once. Banded, half-bandwidth  $N+1$   
  
5 Function k_eigenvalue( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ):  
6    system  $\leftarrow$  BoxSystem( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ) ; // the only factorisation  
7     $\phi_0 \leftarrow \cos(\pi x/2(a/2 + 0.7104))$ ,  $k \leftarrow 1$ ;  
    /* Power iteration (Bell & Glasstone) */  
8    repeat  
9         $q \leftarrow \frac{c \Sigma_t}{k}$  midpoints( $\phi_0$ ), in the  $\phi_0$  row of each cell block ; //  $\phi$  stage  
10        $\Phi \leftarrow$  system.solve( $q$ ) ; // two substitutions, no refactorisation  
11        $\phi_0^{\text{new}} \leftarrow$  moment  $n=0$  of  $\Phi$ ;  
12        $k_{\text{new}} \leftarrow k \frac{\sum \text{midpoints}(\phi_0^{\text{new}})}{\sum \text{midpoints}(\phi_0)}$  ; //  $k$  stage, eq. (25)  
13        $\phi_0 \leftarrow \phi_0^{\text{new}} / \max_j \phi_{0,j}^{\text{new}}$ ,  $k \leftarrow k_{\text{new}}$  ; // rescale: keeps  $k^\ell$  from compounding  
14    until  $|k_{\text{new}} - k| < \epsilon_k |k_{\text{new}}|$ ;  
15    return  $k$   
  
    /* critical_half_thickness, which defers to sn.core */  
16 return critical_size( $a/2 \mapsto k\_eigenvalue(a/2) - 1$ ,  $(a/2)_0$ )
```

Every name in Algorithm 1 is the identifier the code uses. `BoxSystem` is the constructor `pn.box.BoxSystem.__init__`, `splu` is `scipy.sparse.linalg.splu` applied to the CSC form, and `critical_size` is borrowed from `sn.core` - the one place Question 2 reuses the S_N package.

Errors. The box scheme is second order, so the discretisation error in $(a/2)_{\text{crit}}$ falls as Δx^2 and is separable from the truncation error in N by running the same N on a refined mesh - the check reported in Question 3 for S_N .

Results

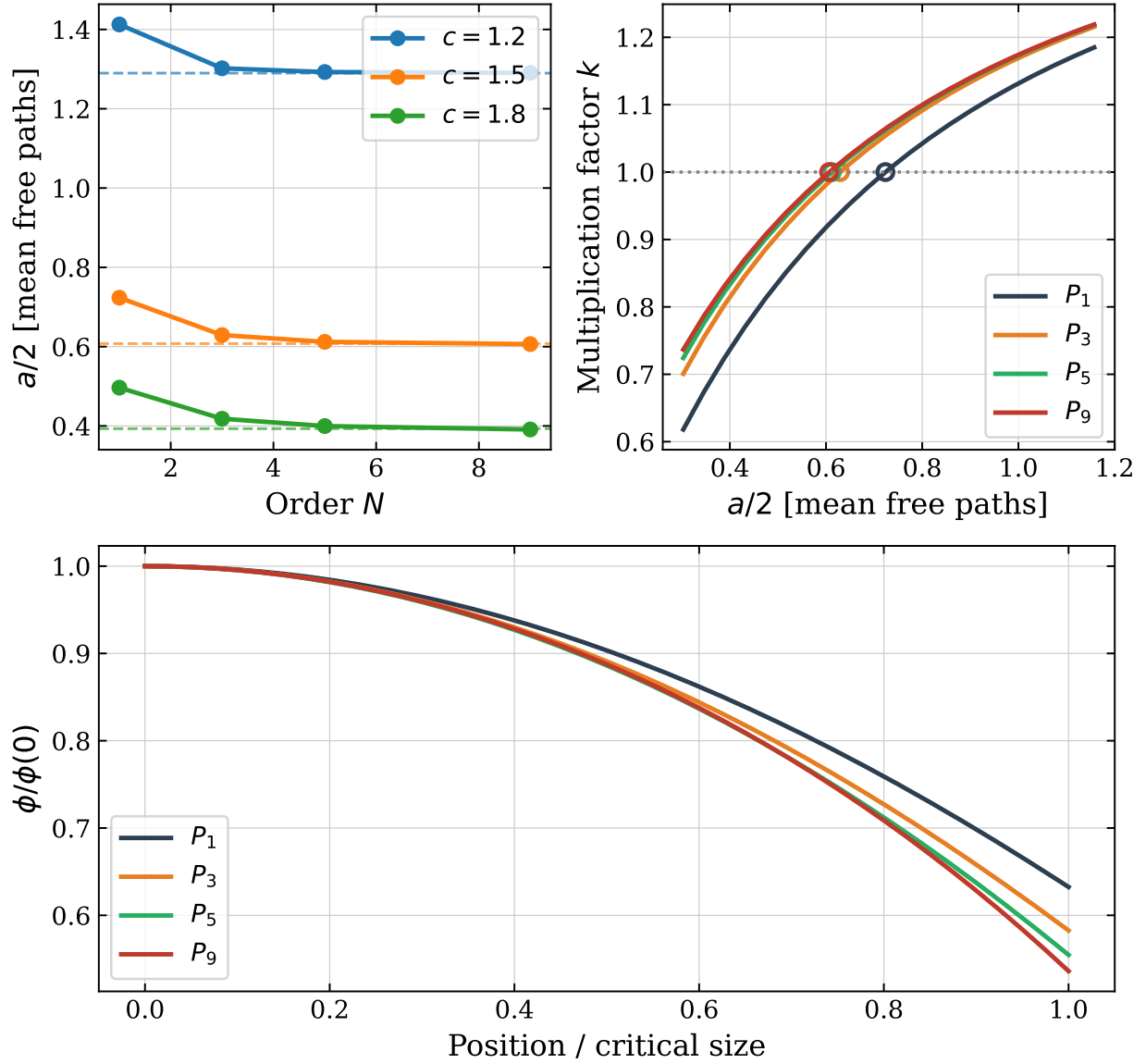


Figure 3: Top left: convergence of $(a/2)_{\text{crit}}$ with N , dashed lines the exact-transport values. Top right: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings, which for P_3 , P_5 and P_9 are already almost on top of one another. Beneath them: the normalised critical flux - as in Question 3, the order changes the surface value and not the interior shape.

c	order	$(a/2)_{\text{crit}}$ [mfp]	exact transport [mfp]	departure
1.2	P_1	1.412502	1.28981	+9.51%
1.2	P_3	1.302001	1.28981	+0.95%
1.2	P_5	1.292597	1.28981	+0.22%
1.2	P_9	1.290382	1.28981	+0.04%
1.5	P_1	0.723480	0.60762	+19.07%
1.5	P_3	0.629149	0.60762	+3.54%
1.5	P_5	0.612136	0.60762	+0.74%
1.5	P_9	0.606467	0.60762	−0.19%
1.8	P_1	0.496560	0.39314	+26.31%
1.8	P_3	0.418210	0.39314	+6.38%
1.8	P_5	0.399736	0.39314	+1.68%
1.8	P_9	0.391130	0.39314	−0.51%

Table 6: Critical half-thickness $(a/2)_{\text{crit}}$ in mean free paths, by the box scheme on 200 cells. The departure is measured against the exact-transport reference $a/2 = \frac{\pi}{2}\nu_0 - z_0$ of Assignment 1.

The S_N method of Questions 3–5

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2} \left[\Sigma_s \phi + \frac{1}{k} \nu \Sigma_f \phi \right], \quad \phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu \quad (28)$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^N w_m \psi_{i,m} \quad (29)$$

Space, slab. The slab equation is (28) at ordinate m , $\mu_m \partial \psi_m / \partial x + \Sigma_t \psi_m = S/2$, whose conservative form integrates over the cell $x \in [x_{i-1/2}, x_{i+1/2}]$ to

$$\mu_m \left(\psi_{i+\frac{1}{2},m} - \psi_{i-\frac{1}{2},m} \right) + \Sigma_t \Delta x \psi_{i,m} = \frac{S_i}{2} \Delta x \quad (30)$$

with the diamond closure $\psi_{i,m} = \frac{1}{2} (\psi_{i+1/2,m} + \psi_{i-1/2,m})$ relating the cell centre to its two faces. Which face is the outflow depends on the sign of μ_m , and that is the whole of the difference between the two sweeps.

Forward sweep, $\mu_m > 0$: the flux enters at $x_{i-1/2}$ and leaves at $x_{i+1/2}$, so the closure is used as $\psi_{i+1/2,m} = 2\psi_{i,m} - \psi_{i-1/2,m}$. Substituting it into (30) removes the outgoing face,

$$\mu_m \left(2\psi_{i,m} - \psi_{i-\frac{1}{2},m} \right) + \Sigma_t \Delta x \psi_{i,m} = \frac{S_i}{2} \Delta x \quad (31)$$

and collecting $\psi_{i,m}$ gives the cell centre and then, from the closure again, the flux handed to the next cell $i+1$:

$$\psi_{i,m} = \frac{\frac{1}{2} S_i \Delta x + 2\mu_m \psi_{i-\frac{1}{2},m}}{\Sigma_t \Delta x + 2\mu_m}, \quad \psi_{i+\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i-\frac{1}{2},m} \quad (32)$$

marched from $x = 0$ outwards, starting from the reflected flux there.

Backward sweep, $\mu_m < 0$: the roles swap. Writing $\mu_m = -|\mu_m|$ turns the leading term of (30) into $|\mu_m|(\psi_{i-1/2,m} - \psi_{i+1/2,m})$, so the flux enters at $x_{i+1/2}$ and leaves at $x_{i-1/2}$, and the closure is used the other way round, $\psi_{i-1/2,m} = 2\psi_{i,m} - \psi_{i+1/2,m}$:

$$\psi_{i,m} = \frac{\frac{1}{2} S_i \Delta x + 2|\mu_m| \psi_{i+\frac{1}{2},m}}{\Sigma_t \Delta x + 2|\mu_m|}, \quad \psi_{i-\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i+\frac{1}{2},m} \quad (33)$$

marched from the vacuum face at $x = a/2$ inwards, starting from $\psi_{I+1/2,m} = 0$. The two are the same formula with the inflow face exchanged, which is why one $|\mu_m|$ serves both.

Space, sphere. The spherical equation carries an angular redistribution term, that can be seen in Eq. (2.46) in Pomraning's book (pg. 30):

$$\mu \frac{\partial \psi}{\partial r} + \frac{1 - \mu^2}{r} \frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \quad (34)$$

whose conservative form integrates over the cell $(r, \mu) \in [r_{i-1/2}, r_{i+1/2}] \times [\mu_{m-1/2}, \mu_{m+1/2}]$ to

$$\mu_m \left(A_{i+1/2} \psi_{i+1/2, m} - A_{i-1/2} \psi_{i-1/2, m} \right) + \beta_{i, m} \left(\alpha_{m+1/2} \psi_{i, m+1/2} - \alpha_{m-1/2} \psi_{i, m-1/2} \right) + \Sigma_t V_i \psi_{i, m} = \frac{S_i}{2} V_i \quad (35)$$

with $A = 4\pi r^2$, $V_i = \frac{4\pi}{3} (r_{i+1/2}^3 - r_{i-1/2}^3)$ and the area increment $\beta_{i, m} = (A_{i+1/2} - A_{i-1/2}) / w_m$. The angular coefficients follow from demanding that a flat flux solve (35) exactly:

$$\alpha_{\frac{1}{2}} = 0, \quad \alpha_{m+1/2} = \alpha_{m-1/2} - w_m \mu_m, \quad \alpha_{N+1/2} = 0 \quad \text{since} \quad \sum_m w_m \mu_m = 0 \quad (36)$$

The recursion needs a starting value $\psi_{i, 1/2}$ at $\mu_{1/2} = -1$, where $1 - \mu^2 = 0$ and the angular derivative drops out. What remains, $-\partial \psi / \partial r + \Sigma_t \psi = S/2$, is (33) at $|\mu| = 1$, run inwards from the vacuum boundary.

The two faces of a spherical cell. Equation (35) is (30) with a second outgoing face. They are not alike and it is worth naming them separately:

- the *radial face*, weighted by the two shell areas $A_{i\pm 1/2}$, whose outflow feeds the **next radial cell** of the same ordinate. Which of the two areas is the outflow depends on the sign of μ_m , exactly as in the slab.
- the *angular face*, weighted by the two coefficients $\alpha_{m\pm 1/2}$ of the angular bin, whose outflow feeds the **next ordinate** in the same cell. This one never swaps: $d\mu/ds \geq 0$ means μ only increases, so the inflow is always at $m - \frac{1}{2}$ and the outflow always at $m + \frac{1}{2}$.

Each closes with its own diamond relation,

$$\psi_{i+1/2, m} = 2\psi_{i, m} - \psi_{i-1/2, m} \quad (\mu_m > 0), \quad \psi_{i-1/2, m} = 2\psi_{i, m} - \psi_{i+1/2, m} \quad (\mu_m < 0) \quad (37)$$

$$\psi_{i, m+1/2} = 2\psi_{i, m} - \psi_{i, m-1/2} \quad (38)$$

Take the forward sweep, $\mu_m > 0$, so that the radial outflow is $A_{\text{out}} = A_{i+1/2}$ and the inflow $A_{\text{in}} = A_{i-1/2}$. Substituting (37) and (38) into (35) eliminates both outgoing fluxes at once:

$$\begin{aligned} \mu_m \left(2A_{\text{out}} \psi_{i, m} - (A_{\text{out}} + A_{\text{in}}) \psi_{i-1/2, m} \right) \\ + \beta_{i, m} \left(2\alpha_{m+1/2} \psi_{i, m} - \left(\alpha_{m+1/2} + \alpha_{m-1/2} \right) \psi_{i, m-1/2} \right) + \Sigma_t V_i \psi_{i, m} = \frac{S_i}{2} V_i \end{aligned} \quad (39)$$

Every term now carries either $\psi_{i, m}$ or a known inflow, so collecting $\psi_{i, m}$ on the left gives the cell solve outright:

$$\boxed{\psi_{i, m} = \frac{\frac{1}{2} S_i V_i + \mu_m (A_{\text{out}} + A_{\text{in}}) \psi_{i-1/2, m} + \beta_{i, m} \left(\alpha_{m+1/2} + \alpha_{m-1/2} \right) \psi_{i, m-1/2}}{\Sigma_t V_i + 2\mu_m A_{\text{out}} + 2\beta_{i, m} \alpha_{m+1/2}}} \quad (40)$$

and (37) and (38) then hand on the two fluxes the march needs next - $\psi_{i+1/2,m}$ to radial cell $i + 1$, and $\psi_{i,m+1/2}$ to ordinate $m + 1$. The backward sweep is (40) with $A_{\text{out}} = A_{i-1/2}$, $A_{\text{in}} = A_{i+1/2}$, μ_m replaced by $|\mu_m|$ and the radial inflow (eq. (37)) taken as $\psi_{i+1/2,m}$. The angular half of the formula is untouched. Setting $A_{\text{out}} = A_{\text{in}} = 1$, $V_i = \Delta x$ and $\alpha_{m\pm 1/2} = 0$ returns (32) exactly, which is the check that the two cell solves are the same balance. **In code they are kept apart on purpose: `sn.slab.cell_flux` takes one inflow, `sn.sphere.cell_flux` takes a radial_face and an angular_face.**

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{\text{in}}$ of (37) or (38) comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. Frankly, this never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t \Delta x / |\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$. Vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu \Sigma_{f,i} \phi_i^{(l)} / k^{(l)}$. Inner iteration n sweeps $S_i = \Sigma_{s,i} \phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles. Then

$$k^{(l+1)} = k^{(l)} \frac{P^{(l+1)}}{P^{(l)}}, \quad P^{(l)} = \sum_i \nu \Sigma_{f,i} \phi_i^{(l)} V_i \quad (41)$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k .

Algorithm 2: One transport solve - backward ordinates, then forward ones -
homework3.sn.slabs / homework3.sn.sphere

Input: isotropic source S_i

Output: scalar flux ϕ_i

Data: $\psi_{\cdot, m-\frac{1}{2}}$, the *angular inflow column* - one value per spatial cell, the half-angle flux ordinate m consumes through its angular face. Sweeping m replaces it with $\psi_{\cdot, m+\frac{1}{2}}$, which ordinate $m+1$ consumes in turn (`psi_low` and `psi_high` in the code). The slab has no such column.

```

1 Function sweep_all_angles( $S$ ):
2    $q_i \leftarrow S_i/2$ ;
3   if sphere then  $\psi_{\cdot, \frac{1}{2}} \leftarrow \text{\_starting\_direction}(q)$  //  $\mu_{1/2} = -1$ : no angular term,
   a plane march;
4    $\phi \leftarrow 0$ .  $\psi_{\frac{1}{2}, \cdot} \leftarrow 0$ ; // the reflective face, one value per ordinate
5   for  $m$  with  $\mu_m < 0$ , ascending do
6      $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}, \psi_{\frac{1}{2}, m'} \leftarrow \text{sweep\_backward}(m, q, \psi_{\cdot, m-\frac{1}{2}})$ ,  $\mu_{m'} = -\mu_m$ ;
7      $\phi \leftarrow \phi + w_m \psi_{\cdot, m}$ ;
8   for  $m$  with  $\mu_m > 0$ , ascending do
9      $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}} \leftarrow \text{sweep\_forward}(m, q, \psi_{\cdot, m-\frac{1}{2}}, \psi_{\frac{1}{2}, m})$ ; // starts from what
    $-\mu_m$  reflected
10     $\phi \leftarrow \phi + w_m \psi_{\cdot, m}$ ;
11  return  $\phi$ 

12 Function sweep_backward( $m, q, \psi_{\cdot, m-\frac{1}{2}}$ ):
13   $\psi_{\text{in}} \leftarrow 0$ ; // vacuum: nothing enters the outer face
14  for  $i \leftarrow I$  to 1 (inwards) do
15    if slab then  $\psi_{i, m}, \psi_{\text{in}} \leftarrow \text{slab.cell\_flux}(\Sigma_t \Delta x, \frac{1}{2} S_i \Delta x, |\mu_m|, \psi_{\text{in}})$  // eq. (33);
16    else  $\psi_{i, m}, \psi_{\text{in}}, \psi_{i, m+\frac{1}{2}} \leftarrow \text{sphere.cell\_flux}(\Sigma_t V_i, \frac{1}{2} S_i V_i, \text{radial\_face}, \text{angular\_face})$ 
   // eq. (40);
17  return  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}, \psi_{\text{in}}$ ; // the last  $\psi_{\text{in}}$  is what reflects at the centre

18 Function sweep_forward( $m, q, \psi_{\cdot, m-\frac{1}{2}}, \psi_{\text{in}}$ ):
19  for  $i \leftarrow 1$  to  $I$  (outwards) do
20    the same cell solve, with the radial inflow and outflow faces exchanged; // eq. (32)
21  return  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}$ 

```

/* The radial face swaps with the sign of μ_m . The angular face never does
- its inflow is always $\psi_{i, m-1/2}$ and its outflow always $\psi_{i, m+1/2}$. */

Every name in Algorithms 2 and 3 is the identifier the code uses. `sweep_backward`, `sweep_forward`, `sweep_all_angles` and `\_starting\_direction` are methods of `sn.slabs.SlabsSolver` and `sn.sphere.SphereSolver` - the two geometries supply them independently, which is why everything in `sn.core` is written against neither. Three nested loops, and not one matrix among them.

Algorithm 3: The two iterations - scattering inside, Bell & Glasstone outside -
homework3.sn.core

Input: medium, geometry, initial guess size₀

Output: critical size

1 **Function** run_sn_for_source(*solver*, *medium*, *Q_f*, *φ*):

```

2   repeat
3   |   φnew ← sweep_all_angles(Σsφ + Qf) ;           // one double sweep per inner
4   |   until maxi |φinew - φi| ≤ εφ maxi φinew or Σs = 0;
5   |   return φ ;                                     // Σs = 0 (Q3, Q4): one sweep is already exact

```

6 **Function** k_eigenvalue(*solver*):

```

7   φ ← 1.  k ← 1.  P ← ∑i νΣf,i φi Vi;
8   repeat
9   |   φnew ← run_sn_for_source(νΣfφ/k) ;           // φ iteration
10  |   Pnew ← ∑i νΣf,i φinew Vi;
11  |   knew ← k Pnew/P ;                             // k iteration
12  |   φ ← φnew/maxi φinew ;                         // rescale, P likewise
13  |   until |knew - k| < εk|knew|;
14  |   return k

```

15 **Function** critical_size(*k of size*, *size₀*):

```

16 |   widen a bracket about size0 by 3% a step until k_eigenvalue() - 1 changes sign;
17 |   return brentq(size ↦ k_eigenvalue(size) - 1)

```

Question 3: Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2}\nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0 .

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 7: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

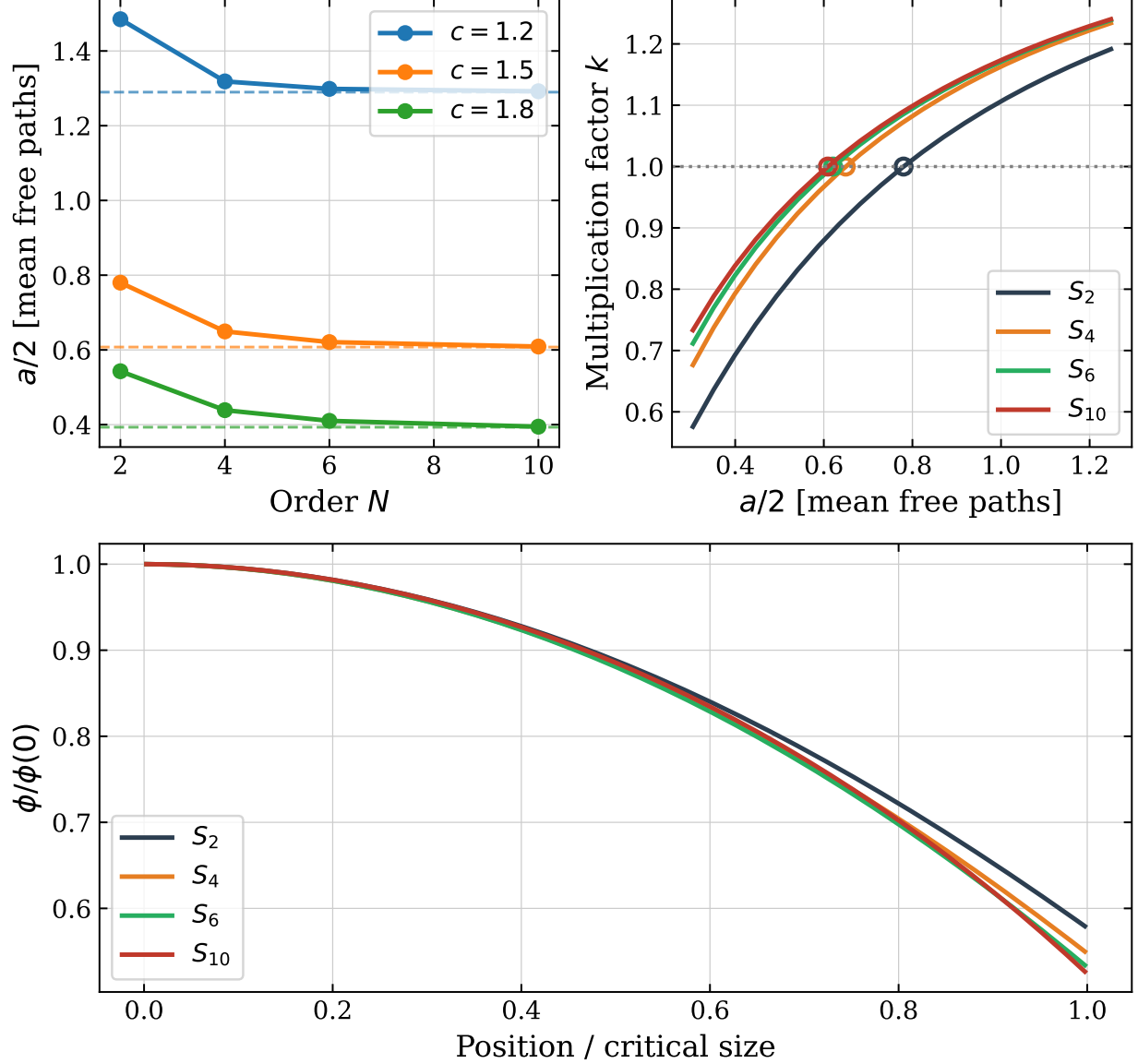


Figure 4: Top left: convergence of $a/2$ with N , dashed lines the exact-transport values. Top right: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Beneath them: the normalised critical flux, which barely moves with N - the order affects the boundary, not the interior.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640$ cm $^{-1}$, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10} , 0.606407 at S_{16} , 0.605631 at S_{24} , 0.605195 at S_{48} .

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss-Legendre points, because a Gauss-Legendre rule of order $N + 1$ integrates polynomials up to

degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate - the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak's*, $\int_0^1 P_{2j+1}(\mu)\psi d\mu = 0$, weight the whole incoming half-range. Mark's pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree. With Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ - the same five digits - while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

c	pair	P_N (Marshak)	S_{N+1} (Mark)	$S_{N+1}/P_N - 1$	exact transport
1.2	P_1 / S_2	1.41250	1.48499	+5.13%	1.28981
1.2	P_3 / S_4	1.30200	1.31855	+1.27%	1.28981
1.2	P_5 / S_6	1.29259	1.29861	+0.47%	1.28981
1.2	P_9 / S_{10}	1.29038	1.29218	+0.14%	1.28981
1.5	P_1 / S_2	0.72348	0.78001	+7.81%	0.60762
1.5	P_3 / S_4	0.62915	0.64949	+3.23%	0.60762
1.5	P_5 / S_6	0.61213	0.62089	+1.43%	0.60762
1.5	P_9 / S_{10}	0.60647	0.60904	+0.42%	0.60762
1.8	P_1 / S_2	0.49656	0.54291	+9.33%	0.39314
1.8	P_3 / S_4	0.41821	0.43875	+4.91%	0.39314
1.8	P_5 / S_6	0.39974	0.41014	+2.60%	0.39314
1.8	P_9 / S_{10}	0.39113	0.39459	+0.88%	0.39314

Table 8: The two methods at matched order, in mean free paths. Every P_N entry is below its S_{N+1} partner and closer to the exact value, and the gap closes as N grows.

Three things in that table follow from the argument above. **The sign of the gap is the sign of l_0** : Marshak's extrapolation $2/3$ is longer than Mark's $1/\sqrt{3} = 0.577$, so the P_N flux is driven to zero further outside the slab and less material is needed - P_N is below S_{N+1} at every one of the twelve entries. **The gap is a boundary effect, so it shrinks with N** , from 5–9% at $N = 1$ to under 1% at $N = 9$. By then both boundary conditions are resolving the same half-range flux and the choice between them stops mattering. **And it grows with c at fixed N** - 5.13%, 7.81%, 9.33% at $N = 1$ - because a larger c means a thinner slab, a shorter relaxation length and a more forward-peaked surface flux, which is exactly the shape a half-range condition of low order cannot capture.

All of that is a statement about *slabs*. In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r} \frac{\partial \psi}{\partial \mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (35) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions.

Question 4: Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (35). The reference is Assignment 1's $\Sigma_t R_c = \pi\nu_0 - z_0$.

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	-3.51%
1.2	S_4	3.14631	3.17204	-0.81%
1.2	S_6	3.15991	3.17204	-0.38%
1.2	S_{10}	3.16728	3.17204	-0.15%
1.5	S_2	1.60725	1.69010	-4.90%
1.5	S_4	1.66739	1.69010	-1.34%
1.5	S_6	1.67934	1.69010	-0.64%
1.5	S_{10}	1.68595	1.69010	-0.25%
1.8	S_2	1.11753	1.18295	-5.53%
1.8	S_4	1.16413	1.18295	-1.59%
1.8	S_6	1.17412	1.18295	-0.75%
1.8	S_{10}	1.17971	1.18295	-0.27%

Table 9: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

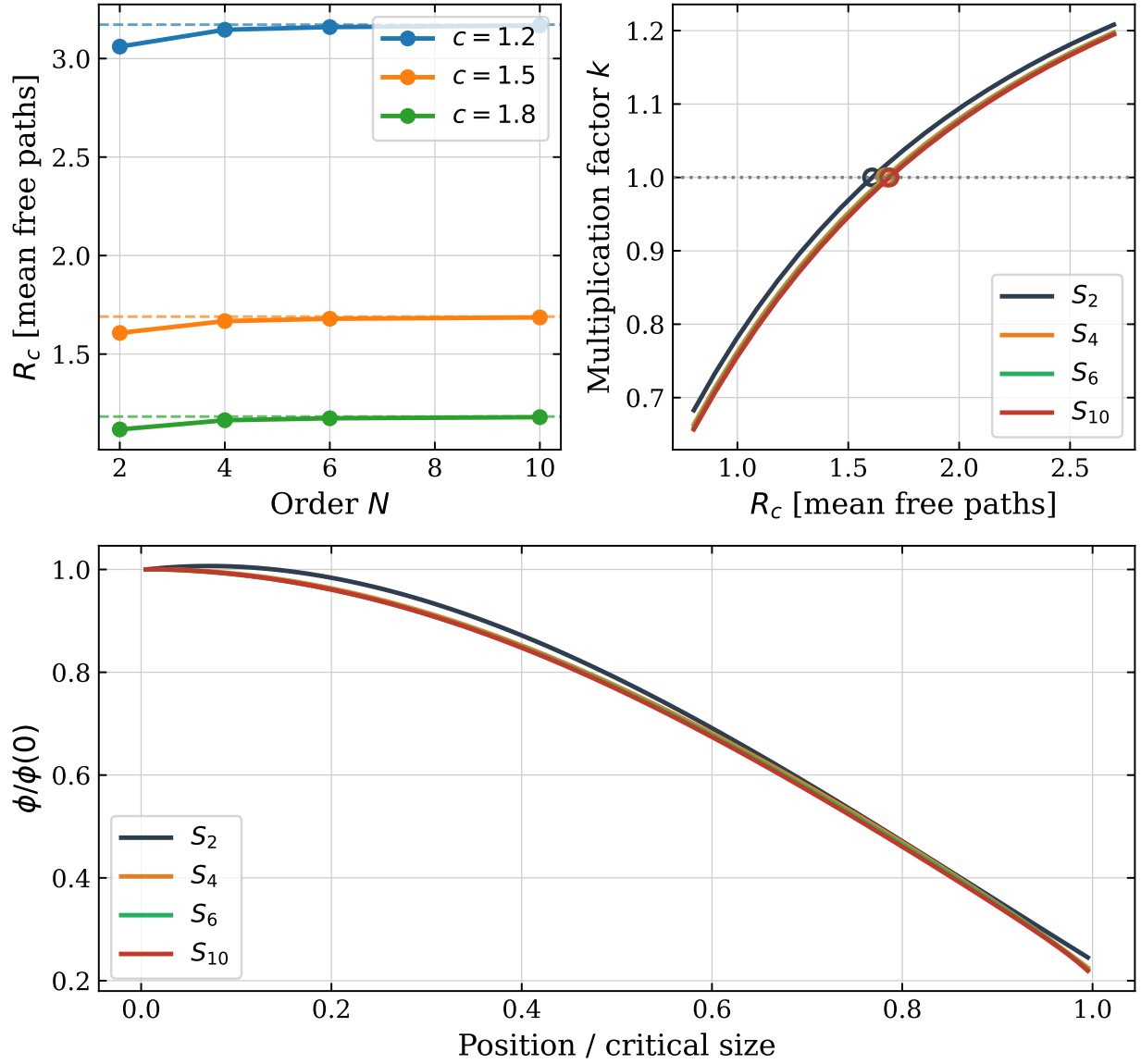


Figure 5: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

Question 5: Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 - $\Sigma_t = 0.32640 \text{ cm}^{-1}$ for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 - and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c -only problems above.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	-5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	-1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	-0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	-0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	-4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	-1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	-0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	-0.15%

Table 10: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

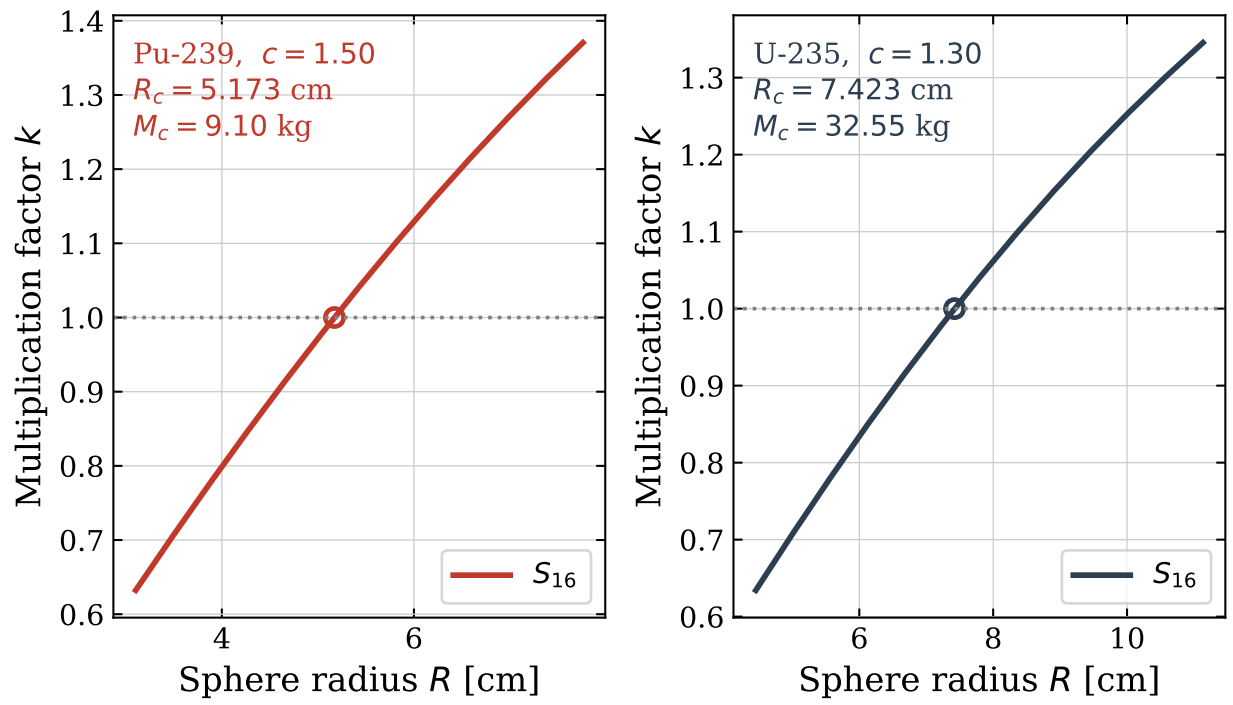


Figure 6: k against sphere radius at S_{16} , with the $k = 1$ crossing that fixes the critical mass.